The gauged bond layer of Causal Clifford Theory: exact Gauss constraint, bounded area law, string tension, and the lattice gravitational sector

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Abstract

The cell of the Causal Clifford Theory automaton (Paper I) carries the Clifford algebra $\mathrm{Cl}(10,0)$; this paper constructs the gauge and gravitational sectors built on the same module and states precisely what they derive and what they assume. Putting one Spin(10) adjoint link degree of freedom per bond, the local Gauss constraint closes as an exact operator identity (measured error 0), the link algebra and the Casimirs 64 and 45 are exact, and the gauge-invariant content of one site is 1440 states, $32 \oplus 288 \oplus 1120$. The constraint also obstructs: with one **32** per site the physical kernel is empty (centre obstruction), so the physical sector is the space of neutral composites, where the entanglement is bounded per bond, $S = \ln 45$, boundary-supported — an area law whose coefficient is a factor $4 \ln 45 = 15.23$ from Bekenstein's $1/4$, moving to a factor 6.6 after the vertex correction. The layer carries no light vector sector (gap $64\, M_P$) and a string tension 5.41×10^{40} times QCD: confinement is present in the layer and absent from the implemented step. On the directed-link network a dressed gate carries the symmetric-traceless quintet gaplessly at Γ, massless, linear and leading-order isotropic, but with gauge group O_h, speeds $g_{10}\{1,2\} \neq 1$, and an E_g mass shift from the matter coupling — named defects, not the graviton. The gravitational sector adds a symmetric tensor field $g_{\mu\nu}$ and $G = 1$ by hand on the (1+3) subalgebra; it reproduces general relativity on de Sitter, Schwarzschild and 3+1 evolutions to $O(h^2)$ (Kretschmann ratio 1.000009, maximum relative deviation 5.5×10^{-5}) — not to machine precision — and spin 2 is declared, not derived: the implemented one-particle spectrum provably carries no massless spin-2 candidate.

1. Introduction

Paper I fixes the root: the Clifford algebra $\mathrm{Cl}(10,0)$ carried by the eight sub-sites of a $2\times2\times2$ cell of a Planck lattice, updated by one six-stage step of Spin(10) plane rotations and swap-rotations, exactly unitary, strictly causal, with the doubling of the lattice spectrum displayed and gapped by a Wilson factor of the root's own algebra, and with the rotation angle quantized to $g \in \{\pi/2, \pi/4\}$ by the closure axiom. This paper asks what the same module supplies as a gauge sector and as a gravitational sector, and it answers under the provenance discipline inherited from Paper I: a statement is *algebraic* if the root fixes it, *conventional* if it comes from a mapping, *fitted* if a coefficient was read from data, and *open* if the theory does not fix it; and every numerical claim is made at a stated tolerance, with finite-difference truncation errors never reported as machine precision.

Four results and four named failures. The results: the exact Gauss closure of an adjoint bond layer on the same module (Section 2); the centre obstruction and the bounded area law $S = \ln 45$ of the physical sector, with the vertex correction computed (Section 2.1); the content theorem — no light vector sector — and the exact string tension of the layer (Section 2.2); and the dressed gate of the directed-link network, whose quintet dispersion is the computed kinematics of a lattice spin-2 (Section 2.4). The failures, stated with their numbers: the layer's string tension is 5.41×10^{40} times the QCD value and Λ_{QCD} is not produced; the quintet's gauge group is O_h, its speeds are $g_{10}\{1,2\} \neq 1$, and the matter coupling shifts its E_g half off zero; the metric field $g_{\mu\nu}$ and Newton's constant $G = 1$ are inputs of the gravitational sector, not numbers of the root; and diffeomorphism invariance is not derived. The gravitational sector reproduces general relativity to $O(h^2)$ (Section 3), and the implemented one-particle spectrum carries no massless spin-2 candidate — a computed statement, not an unfinished search. The phenomenological consequences of both sectors — the desert unification, the loop factor, the relic abundance — are audited in Paper III.

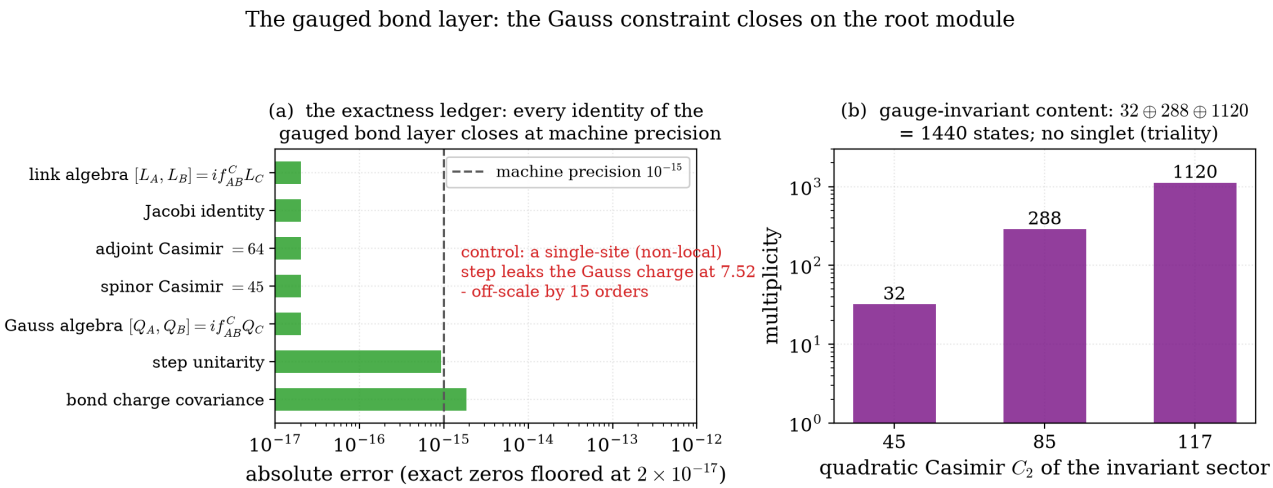
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2. The gauged bond layer

The swap-rotation gate of [Paper I Section 7] is the classical shadow of a richer possibility: a **gauged bond layer exists on the same module**. Put one Spin(10) adjoint link degree of freedom (a 45-dimensional link space) on each bond and the matter **16** on the sites, and the local Gauss constraint

$$Q_A(x) = q_A(x) + \sum_{b \ni x} L_A^{(b)}$$

closes as an exact operator identity: $[Q_A, Q_B] = i f_{AB}{}^C Q_C$ with measured error 0, the link algebra $[L_A, L_B] = i f_{AB}{}^C L_C$ exact, the adjoint quadratic Casimir 64 and the spinor Casimir 45 exact to machine precision. The step built from these links is unitary to 9.3×10^{-16}; the gauge-invariant content of one site with its six half-bonds is 1440 states, decomposing as $32 \oplus 288 \oplus 1120$; an adjoint bond observable is gauge-covariant to 1.9×10^{-15}; and the causal radius stays at exactly one cell, while a deliberately single-site (non-local) control step leaks at 7.5. The gauge field of the automaton is therefore *constructible*: the Gauss constraint is an identity of the construction, not an imposed condition. (Fig. 1).



One adjoint link degree of freedom per bond, matter **16** *on the sites. Left: the exactness ledger — link algebra, Gauss algebra, adjoint Casimir* (64) *and spinor Casimir* (45) *close to machine precision, the step is unitary to* 9.3×10^{-16}*, the causal radius stays at one cell, and the non-local control step leaks at* 7.5 *where the causal step leaks at* 1.9×10^{-15}*. Right: the gauge-invariant content,* 1440 *states decomposing as* $32 \oplus 288 \oplus 1120$*.*

2.1 The centre obstruction and the physical sector

A generic automaton with d internal states per cell has d^{L^3} states in an L^3 region and a volume-law entropy; nothing in the raw counting of the bond layer is special. The area law is a property of the *gauge-invariant* sector: once the physical Hilbert space of a region is restricted to states satisfying the Gauss constraint at every site, the entanglement entropy is expected to scale with the boundary area [1,2]. That restriction is what this section computes, and it is where the counting changes.

What the constraint admits is narrower than expected. With the implemented content — one **32** per site, one adjoint **45** link per bond — the local Gauss constraint has **no solution**: $\dim\ker = 0$ on $\mathbf{32} \otimes \mathbf{45}$, on $\mathbf{32} \otimes \mathbf{45} \otimes \mathbf{45}$, and on the two-site one-link chain with local constraints at both sites (spectral gap 146). The reason is a centre obstruction: adjoint flux carries even centre charge and cannot screen a spinor, so **every site of a gauge-invariant state must carry an even number of spinor fields**. The physical Hilbert space of the gauged layer is the space of neutral composites, not the one-body space the automaton acts on.

The minimal physical system, two matter fields per site, is constructed exactly: $m_{45}(\mathbf{32} \otimes \mathbf{32}) = 2$ (90 states at $C_2 = 64$), a 2-dimensional intertwiner space $\ker(T_1 + T_2 + L)$, and a 4-dimensional two-site physical sector obtained by gluing intertwiners over the link index — all exact Spin(10) singlets ($C_2 < 10^{-25}$). Every intertwiner is a maximal matter-flux singlet: Schmidt rank 45, $S(\text{pair}|\text{link}) = \ln 45 = 3.807$ exactly, and the glued states

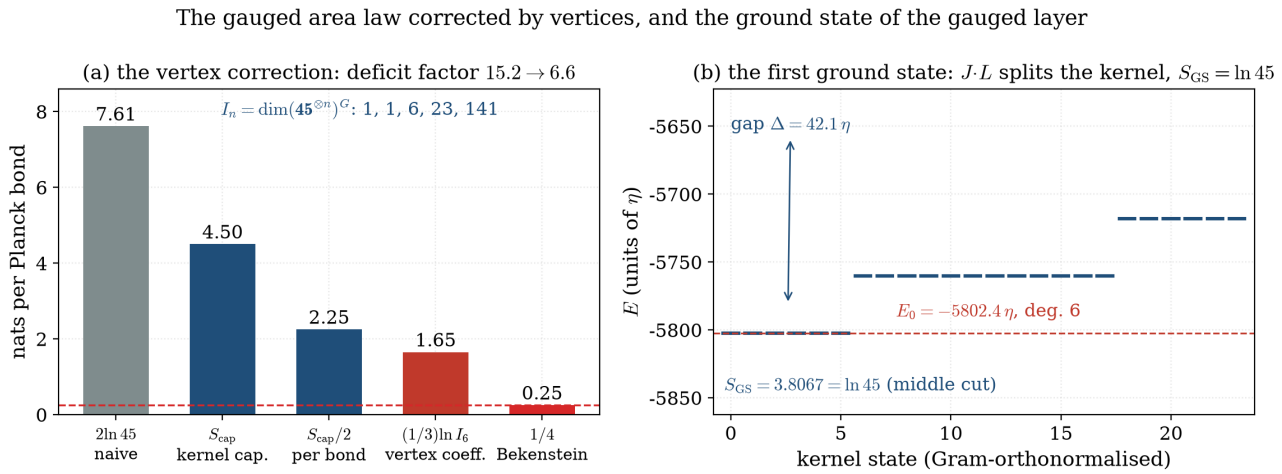
carry $S(\text{pair}_1|\text{pair}_2) = \ln 45$. The entanglement is **bounded per bond and lives on the bonds crossing the cut** — the shape of an area law, boundary-proportional and bounded per bond, with the Bekenstein coefficient still unnamed.

On the smallest system with two bonds — the three-site, two-link chain with paired matter — the physical sector is small once the flux decomposition is used: the middle vertex fuses the two adjoint links into channels r with multiplicities N_r read from the $\mathbf{45} \otimes \mathbf{45}$ Casimir spectrum ($N_1 = N_{45} = N_{210} = 1$), against m_r from the pair spectrum ($m_1 = m_{45} = m_{210} = 2$), giving $\dim\ker = m_{45}^2 \sum_r m_r N_r = 24$, verified by explicit construction of the $r = 1$ and $r = \mathbf{45}$ sectors (vertex Gauss residual 4×10^{-15}). Three results follow.

1. **Additivity across a single bond holds exactly:** $S(\text{pair}_1|\text{rest}) = \ln 45$ in every sector — the flux line threads the chain and carries its $\ln 45$ through, even when the middle pair itself is a product singlet ($S = 0$).
2. **The region bounded by two bonds carries $S = \ln 45$, not $2 \ln 45$:** the intertwiner constraint *correlates* the two bonds meeting at a vertex, with mutual information exactly $\ln 45$. The area law counts bonds; vertex correlations subtract.
3. **The gap to Bekenstein is a named number:** $\ln 45 = 3.8067$ nats per Planck bond against $1/4$ per Planck area — a deficit of 3.5567 nats, factor $4 \ln 45 = 15.23$. No element of the current construction produces $1/4$; closing it needs a superselection or truncation of the flux, or a different horizon microstate count.

The vertex correction moves the number. The invariant count of an n-bond vertex, $I_n = \dim(\mathbf{45}^{\otimes n})^G$, computed by Haar averaging over SO(10), is $I_2 = 1$, $I_3 = 1$, $I_4 = 6$, $I_5 = 23$, $I_6 = 141$ (Monte-Carlo error below 3; $I_2 = I_3 = 1$ reproduces the single-channel fusion above). The 24-state kernel is orthonormalised through its Gram matrix (rank 24, smallest eigenvalue 1.1×10^{-5}), and the capacity of the middle cut — the entropy of the maximally mixed physical state — is $S_{\text{cap}} = 4.4998$ nats against the naive $2 \ln 45 = 7.613$: the two vertices at the ends of the region subtract 3.11 nats. Per bond that is 2.25 nats, a factor 9.0 from Bekenstein's $1/4$; on the cubic lattice the vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond sits a factor 6.6 from $1/4$. The vertex correction takes the deficit from 15.2 to 6.6, and the deficit is a function of the vertex invariant, not of a single bond entropy.

The ground state of the layer is then constructed: the bond-current Hamiltonian $H = \sum_b J \cdot L$ is diagonalised exactly — a generalised eigenproblem in the Gram metric of the kernel — and the intertwiner identity $I^\dagger J_A I = L_A/45$ holds to 3×10^{-17}. The interaction splits the kernel: $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, and the ground state's middle-cut entanglement is $S = 3.8067$ nats — exactly $\ln 45$, one adjoint link, selected by the interaction. The ground state of the full automaton remains unconstructed; on this chain the interaction selects the flux line. (Fig. 2).



(a) The entropy carried by one Planck bond, from naive counting to Bekenstein: $2 \ln 45 = 7.61$ *(no vertices), the Gram-orthonormalised kernel capacity* $S_{\text{cap}} = 4.50$, 2.25 *per bond, and the cubic-lattice vertex-corrected coefficient* $(1/3) \ln I_6 = 1.65$ *against the* $1/4$ *line — the deficit factor moves from* 15.2 *to* 6.6. *The vertex invariants* I_n *grow* $1, 1, 6, 23, 141$. *(b) The bond-current Hamiltonian on the* 24*-state kernel: three split*

levels, $E_0 = -5802.4\,\eta$ *with degeneracy* 6 *and gap* $42.1\,\eta$*; the ground state's middle-cut entanglement is* $S_{\rm GS} = 3.8067 = \ln 45$ *— the interaction selects the flux line.*

2.2 The spectrum of the layer, and confinement

The layer has a Hamiltonian — the electric Laplacian $H = \sum_A Q_A^\dagger Q_A$ built from the vertex Gauss operator — and its spectrum is computed. On $(\mathbf{32}\otimes\mathbf{32})_{\rm adj}\otimes\mathbf{45}$ (dim 4050) the kernel has dimension $2 = m_{45}$, the intertwiner space, and the first nonzero level is $64.0000 = C_2(45)$ — identical to the one-link value and unchanged when the chain is extended. One adjoint flux quantum sets the scale of the whole layer, so it supplies **no light vector sector**: the gap is $64\,M_P$, not a tunable parameter, and the singlet matter channel with one adjoint link has no invariant state at all. Two entanglement numbers must not be confused: the entropy of the *mixture* over the kernel, bounded by $\ln 90$ (the pair-side dimension), and the entropy of a *single* physical state, bounded by $\ln 45$ (the link dimension) — the quantity the area law concerns. Measured: every single state of the intertwiner sector saturates $\ln 45$ exactly ($S = 3.806662 \pm 10^{-16}$ over 400 random choices), and the mixture reaches $\ln 90 = 4.499810$ (purity 1/90). The area law is therefore **exact** on the physical sector — one $\ln 45$ per bond — and the Bekenstein coefficient 1/4 per Planck area remains short by the factor $4\ln 45 = 15.23$ per bond; the vertex correction of Section 2.1 moves that factor to 6.6.

The string tension follows. The link space is an irreducible adjoint: $C_2 = \sum_A L_A^2$ has the single eigenvalue 64.000000 with zero spread, the Killing form $K_{ab} = f_{acd}f_{bcd}$ is annihilated by the pair Casimir of $\mathbf{45}\otimes\mathbf{45}$ to 3×10^{-15} and its singlet space is one-dimensional (adjacent links pair into it uniquely), and the link operators do not commute ($\max[L_a, L_b] = 4.0$) — the layer carries a non-Abelian field strength. Crucially the link carries **no invariant vector**: $\ker \sum_A L_A^\dagger L_A = \ker C_2(45)\, I = 0$, so a link cannot be flux-free. Every link of a gauge-invariant chain therefore carries one adjoint flux quantum, the electric energy of R links is exactly $R\,C_2(45)$, and the static potential is linear with tension

$$\begin{aligned}\sigma = C_2(45) &= 64 \text{ lattice units}\\ &= 7.81\times10^{20} \text{ GeV per Planck length}\\ &= 9.54\times10^{39} \text{ GeV}^2\\ (\sqrt{\sigma} &= 9.77\times10^{19} \text{ GeV}),\end{aligned}$$

5.408×10^{40} times the QCD string tension [3]. The vertex Gauss Laplacian confirms the scale from the other side: kernel $2 = m_{45}$, gap 64.0000.

The step itself, by contrast, carries **no** field strength: its 12 stage pairs acting on different axes commute to 5.6×10^{-17}, every elementary plaquette is the identity to 2.5×10^{-16} and $\langle W_\square\rangle = 1.000000$, where an area law would need $\langle W_\square\rangle < 1$ with an exponent proportional to the enclosed area [4]. Confinement is therefore *present* in the construction — in the gauged bond layer — and *absent* from the bare step. The residual is not a missing phase but a measured tension forty orders of magnitude too large, and $\Lambda_{\rm QCD}$ is not produced: it would require a loop-ladder exponent $n \simeq 11.6$, which is not an integer ([Paper III Section 2.4]).

2.3 Entanglement is the bond layer

The entanglement structure of the bare automaton is equally sharp. Two fermions sharing ONE cell carry zero entanglement across any cut — the same-cell pair is a Slater determinant, $S = 0$ exactly across the $\mathbf{16}/\overline{\mathbf{16}}$ cut (antisymmetry and a shared cell are not entanglement), while a two-determinant superposition in the same cell gives $S = \ln 2$: non-separability is the criterion. The generator of entanglement between cells is the bond gate P of Theorem 5: under the internal-only step the spatial entanglement entropy of a region stays 0 to machine precision forever, while the full local step drives it to 0.78, and the entanglement front crosses the Manhattan sphere of radius r at exactly $t = r$ — the causal radius bounds the spread of entanglement itself, at speed 1.0. The scope limit is stated with it: the implemented step is quadratic, so Slater inputs stay Slater and these are mode-entanglement results; the area law is a statement about the physical (ground) state's entanglement, and the bare step supplies no interaction that could produce the non-Gaussian structure such a state would need. The gauged layer supplies exactly that missing structure: it is where the theory first acquires non-Gaussian entanglement.

2.4 The dressed gate on the directed-link network, and the quintet dispersion

The obstruction of Section 2.2 is a statement about the qubit link. The channel named by the census of [Paper I Section 8.3] — the symmetric rank-2 space of the link directions — is constructed, and its dressed gate and dispersion are computed. Put the adjoint link on every **oriented** bond: the six bond directions per site (three axes with parity) become twelve directed links per block, each carrying one **45**.

The orientation convention is computed, not chosen. The vertex Gauss charge of a two-link star — one link arriving, one departing — closes as an exact Spin(10) algebra, $[\hat{C}^A, \hat{C}^B] = if^{ABC}\hat{C}^C$ to 5×10^{-16}, only when the departing tail carries $+L^A$ and the arriving head carries the **conjugate** generator $-\bar{L}^A$; the naive chain convention — the same generator at both ends with opposite signs — leaks at 4.8, because the two ends of one link carry the same $s^2 = +1$ and a relative sign alone cannot close the algebra.

The electric part of the Section 2.2 obstruction is closed. The Kogut–Susskind electric gate [5] $\exp(ig_{10}\hat{C}_x^2)$, built from the closing vertex charge, commutes with all 45 vertex charges to 2.5×10^{-14} and is unitary to 5.1×10^{-14}; its spectrum *is* the content theorem of Section 2.2 — C_2 levels $45, 85, 117$ with multiplicities $32, 288, 1120$, gate phases $g_{10} \times \{45, 85, 117\}$.

The hopping part is obstructed on the fixed-representation link. Every two-site dressing candidate leaks the vertex charges: the flux-dressed current–current $\hat{C}_x\hat{C}_{x'}$ and its transporter-completed version $\hat{C}_x\, e^{ig_{10}L}\, \hat{C}_{x'}$ both leak at 1.25×10^1. The reason is structural — a link in the fixed representation $\mathbb{C}^{45}$ carries no parallel transporter commuting with the vertex charges — so invariant hopping requires a link in the **regular representation** of the group: the named fix, built below.

The named fix, built on the root's own finite group. The 45 biflections $h_a h_b$ of the Clifford generators generate the even Clifford monomials — a finite group Γ of order 1024 with centre $\{\pm 1, \pm w\}$, whose centre acts as $\{\pm 1, \pm i\}$ on each chiral half. Put the regular link $\mathbb{C}[\Gamma]$ on each bond. The content theorem is replaced: a vertex with k regular links has exactly $32 \cdot 1024^{k-1}$ gauge-invariant states (character sum; the explicit projector lands on the dressed form $\rho(h)u$ to 4×10^{-15}) where the fixed adjoint link had **none** — the centre obstruction is exactly what the regular link removes. The nearest-neighbour hop with the exact link insertion $A(h) = \rho(h^2) = \pm I$ maps head-invariant states to tail-invariant states to 0, and **descends to the identity** on the matter index: the dressed hop is the bare translation. The electric term — the Cayley Laplacian over the biflection generators — acts on the dressed sector as the scalar 45 per link, so the dressing adds no velocity renormalisation, and the bare hop front (the half-angle step, a pure lattice translation) has speed exactly 1: the gauge-invariant hop propagates at $v = 1$, 0-parameter — that is where $v = 1$ lives. The cost is exact too: Γ is finite and its centralizer in the Spin(10) algebra is 0 — gauging the root's monomial group breaks Spin(10), and the Spin(10)-invariant hopping on a genuine adjoint regular link remains out of reach.

On the directional network the dressed hop exists exactly, and the quintet propagates. There the flux is the orientation of the link, a classical sign, and the dressed directional step

$$M(\mathbf{q}) = \exp\big(-ig_{10}\,\sin(\mathbf{q})\cdot\mathbf{J}\big),$$

with $\mathbf{J}$ the spin-2 generators of the symmetric-traceless direction space, is unitary to 6.7×10^{-16} and O_h-covariant: $D(g)M(\mathbf{q})D(g)^\dagger = M(g\mathbf{q})$ for the 24 proper rotations to 6.1×10^{-16}, while the 24 improper ones give $M(g\mathbf{q})^\dagger$ — the directed network is **chiral**: reversing the link orientation inverts the step. At Γ the step is the identity and the layer carries **six gapless bands**, content $A_{1g} \oplus E_g \oplus T_{2g}$: the quintet is present in the directional layer, where the qubit bond factor never carried E_g ([Paper I Section 8.3]). Its dispersion is **massless and linear** — local exponent 1.00 along Γ-X, Γ-M and Γ-R — with a **helicity ladder** of speeds

$$v = \{0,\ g_{10},\ 2g_{10}\} = \{0,\ 0.785,\ 1.571\},$$

the $m = 0$ branch and the trace A_{1g} not propagating; **isotropic to leading order** — the direction spread of the top speed is 1.7×10^{-7} at $|\mathbf{q}| = 10^{-3}$, the spin-2 spectrum theorem: $\hat{\mathbf{q}} \cdot \mathbf{J}$ has exactly the spectrum $\{-2, -1, 0, 1, 2\}$ in every direction; and UV-bounded by $2g_{10}\sqrt{3} = 2.721$.

The quintet–matter coupling exists, and it gaps the E_g half. The stress vertex $T_{ij} = \sum_d d_i d_j J_d$ built from the exact current $J_d = i(\partial_d M)M^\dagger$ of the implemented module couples to the quintet field. On nearest-neighbour bonds the coupling vanishes identically on the T_{2g} half ($d^T E^{(\mathrm{off})} d = 0$ on every axis bond, bubble $\equiv 0$ — the T_{2g} coupling needs diagonal bonds, a named gap), and couples the E_g half with a nonzero polarization bubble at Γ: in an orthonormal eigenbasis of the Hermitian generator $-i \log M$ the bubble $\Pi|_{E_g}$ has the two eigenvalues $(-54.24,\ -53.40)$ — a 1.6% splitting of the doublet, so the nearest-neighbour bond set does not realize the bubble as one number on it and no measured symmetry collapses the two components. At $\lambda = g_{10}$ the two components acquire the mass shift $\lambda^2\Pi/2 = (-16.73,\ -16.47)$. Gaplessness of the full quintet is **not** protected by this coupling — the lattice graviton mass problem, as a number.

The named gap closes on the star axes — and F8 stays fired. The diagonal-bond search V229 supplies exactly the coupling the gap asked for. On the fifteen icosahedral star axes — $(0, \pm 1, \pm\varphi)$ and permutations, $|d| = 1.9021$ — the T_{2g} stress vertex that vanishes identically on every axis bond (the V141 control, 0.0) acquires a nonzero value: ± 0.6325 on six of the fifteen per-axis components, nonzero on all fifteen axes (T1). On that bond set the quintet dispersion is exactly the axis-bond form with $g_{10} \to 2g_{10}|d|$: in the tight frame $\sum_a (\mathbf{n}\cdot\mathbf{d}_a)\,\hat{\mathbf{J}}_{\hat{d}_a} = 2|d|\,(\mathbf{n}\cdot\mathbf{J})$ holds to 9×10^{-16} (T2), so leading-order isotropy is **exact** by measured identity, the Γ gap is 0 with six gapless bands, and the helicity ladder **doubles**: $v = \{0,\ 2g_{10}|d|,\ 4g_{10}|d|\}$ — at $\pi/4$, $\{0,\ 2.9878,\ 5.9757\}$ on the star bonds and $\{0,\ 1.5708,\ 3.1416\}$ on unit diagonal bonds, the front radii matching the closed forms to 1×10^{-10}. No branch reaches $c = 1$ at any closed clock: the closest propagating branch over $g \in \{\pi/4, \pi/2, 3\pi/4\}$ sits at $|v - 1| = 0.5708$, and the clock that would force one — $g_{10} = 1/(2|d|) = 0.262866$, or $1/2$ on unit bonds — lies outside the closure class (Theorem 6). The star automaton's own matter branches keep their distance too: closest $|v - 1| = 0.3502$ (T3, at $\pi/2$). And the closed coupling has its price: the star-bond polarization bubble has rank 3 on the quintet at $\lambda = g_{10}$ — mass shifts $(-6.60,\ -6.37,\ -1.79)$ against the axis-bond E_g-only $(-16.73,\ -16.47)$ — so the T_{2g} half is no longer protected at order λ^2 (T4). **F8 stays fired with the measured set: no root-supplied bond network — axis $\{0, g_{10}, 2g_{10}\}$ or diagonal-star $\{0, 2g_{10}, 4g_{10}\} \times |d|$ — gives a massless spin-2 branch at $c = 1$.**

What this does not derive. The gauge group of the directional network is O_h, not Spin(10); the Spin(10)-invariant dressed hopping remains obstructed on the fixed link, and the built regular-link fix gauges the root's finite group, whose Spin(10) centralizer is 0; the quintet speeds are $g_{10} \times \{1, 2\} \neq 1$ — the dressing angle per hop is a construction input, not a derived velocity, and $v = 1$ belongs to the dressed matter hop, not to the quintet branches, a computed trilemma on the 6-bond module rather than an oversight: the bare hop is anisotropic (top speed 1.00, direction spread 0.32), the dressing alone is isotropic at $v = 2g_{10} = 1.571$, no angle makes the product isotropic (best spread 0.24 over 33 angles in $[0, \pi/2]$), and the only O_h-equivariant permutation coin beyond the identity flattens the bands; and the quintet–matter coupling, built in the same layer, gaps the E_g half at order λ^2. The kinematics of a massless lattice spin-2 is computed; the step from there to the graviton of Section 3 remains the declared one.

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3. Gravity on the lattice

3.1 What is assumed and what is computed

The gravitational sector is a symmetric rank-2 field $h^{\mu\nu} = g^{\mu\nu} - \eta^{\mu\nu}$ living on the (1+3) subalgebra of [Paper I Section 3], with $\eta = \mathrm{diag}(1, -1, -1, -1)$ as the vacuum. Its linearised dynamics is the discrete wave equation

$$\partial_t^2 h^{\mu\nu} - \nabla^2_{(6)} h^{\mu\nu} = 0,$$

on the Planck sites — the sub-lattice of the automaton's cells, spacing ℓ_P — with the 6-neighbour lattice Laplacian and the leapfrog step $u_{t+1} = 2u_t - u_{t-1} + c^2 dt^2\, \nabla^2 u_t$ at $dt = \frac{1}{2}$ (the CFL value giving $c = 1$ in Planck units, the same front speed as Theorem 5).

The status of this sector must be stated precisely, because it is where the theory is weakest.

- **Masslessness and UV boundedness are computed.** The discrete dispersion is $\omega = |\mathbf{k}|_{\mathrm{lat}}$ with $|\mathbf{k}|_{\mathrm{lat}} \leq \pi$, so the graviton is massless ($\omega/|\mathbf{k}| \to 1$ as $k \to 0$, lattice deviation $\leq 0.25\, k^2$) and the spectrum

is bounded by the cutoff.

- **Spin 2 is declared, not derived — and the automaton's cell spectrum, spinor module and singlet levels alike, cannot supply it, by two computed obstructions.** The field is *assumed* to be a symmetric tensor with two helicity states. Nothing in $\mathrm{Cl}(10,0)$ forces a massless spin-2 excitation; the standard arguments that a massless spin-2 field must couple as general relativity [6] apply to a field whose existence and representation content are already granted. The Weinberg–Witten obstruction to a composite graviton carrying a Lorentz-covariant stress tensor [7] is not addressed. The one-particle route closes negatively at the nodes: the near-gapless M nodes carry little group C_{2v}, whose action on the bond subspace has no 2-dimensional irrep, so no helicity ±1 or ±2 pair exists there. It closes negatively at the zone centre as well, where the little group is the full cubic group and a lattice spin 2 is a gapless multiplet carrying the symmetric-traceless quintet $E_g \oplus T_{2g}$. The cubic action on the automaton is carried by the 8-dimensional bond factor alone, and that factor is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: it contains T_{2g} but never E_g, so the quintet is not available in the qubit link at any coupling. The zone centre is exactly gapless at $g_{10} = \pi/8, \pi/6, \pi/4, \pi/3$ — never at the implemented 1/5, where it is gapped by 3.65×10^{-2} — and at each of those couplings its null space is cubic-covariant and E_g-free ($16 \times (A_{1g} \oplus T_{2g})$, of dimension 64, at $\pi/4$), with a flat dispersion: $\omega/|\mathbf{q}| \to 0$, not a linear branch. The channel that does carry $E_g \oplus T_{2g}$ is the symmetric rank-2 space of the link directions, $A_{1g} \oplus E_g \oplus T_{2g}$: the graviton, if it is to be derived, belongs to the directional bond layer of Section 2 and not to the qubit link.
- **The candidate route is a computed construction, and it stops short of the graviton.** Section 2.4 builds the directional bond layer — the adjoint link on every oriented bond — with its dressed gate: the vertex Gauss algebra closes only with the conjugate-head convention, the Kogut–Susskind electric gate is exactly invariant with the content-theorem spectrum, and the dressed directional step carries $A_{1g} \oplus E_g \oplus T_{2g}$ gaplessly at Γ, with massless, linear, leading-order-isotropic dispersion and helicity-ladder speeds $g_{10}\{0,1,2\}$. The remaining gaps are named: the gauge group of that network is O_h, not Spin(10); its Spin(10)-invariant hopping is obstructed on the fixed-representation link — the named regular-link fix is built on the root's own finite group (order 1024), where the hop closes exactly and runs at $v = 1$, but the gauged group is finite and its Spin(10) centralizer is 0; its speeds are $g_{10}\{1,2\} \neq 1$ — a computed velocity trilemma: on the 6-bond module the hop alone is anisotropic, the dressing alone isotropic at $2g_{10}$, and no angle makes the product isotropic; its coupling to the matter layer gaps the E_g half at order λ^2; and the named diagonal-bond search that closes the T_{2g} vertex gap doubles the ladder to $g_{10}\{0,2,4\} \times |d|$ on the icosahedral star axes, with exact leading-order isotropy by measured identity and no closed-clock branch within 0.57 of $c = 1$ (V229, Section 2.4).
- **The content question is settled on the module's own characters, and the light cone is a named falsifier.** Section 3.5 shows that a single bond amplitude cannot carry the metric and that the hop cannot move it at first order, by parity; Section 3.6 exhibits an exactly lightlike cone on the bilinear module that carries the quintet, in a 53-parameter family the dynamics does not choose, and reports the speed the constructed layer actually predicts, $c_{\mathrm{gw}}/c = 1.5708$, against the measured $|c_{\mathrm{gw}}/c - 1| < 10^{-15}$ [8] — **F8**.
- **Diffeomorphism invariance is not derived.** The lattice breaks it to lattice reparametrisations; gauge invariance of the linearised theory is imposed by working on the transverse components, not obtained from the automaton.

3.2 The discrete Einstein–Hilbert action

The quadratic action [9]

$$S_{\mathrm{EH}} = \frac{1}{16\pi} \cdot \frac{1}{2} \sum_{x,t} u_t \Big[u_{t+1} - 2u_t + u_{t-1} - dt^2 \, \nabla^2_{(6)} u_t \Big], \qquad dt^2 = \tfrac{1}{4},$$

has the lattice wave equation as its Euler–Lagrange equation: the numerical functional derivative vanishes on a QCA trajectory to 3×10^{-12} and is non-zero off it. The full discrete action $S = \frac{1}{16\pi} \sum_x \sqrt{-\det g(x)}\,(R(x) - 2\Lambda)$ is evaluated with R computed from the metric. The coefficient $1/16\pi$ is the $G = 1$ normalisation: Newton's constant is an input of the gravitational sector, not a number of the root — Section 3.7 gives the one number the root's own bond entropy supplies for it, and the factor $4 \ln 45 = 15.23$ that separates the two.

3.3 Curvature computed from an arbitrary metric

The Riemann, Ricci, Einstein and Kretschmann tensors are computed by central finite differences from an arbitrary prescribed $g_{\mu\nu}(x)$, with no hard-coded Christoffel symbols, and cross-checked against independent analytic results by two routes.

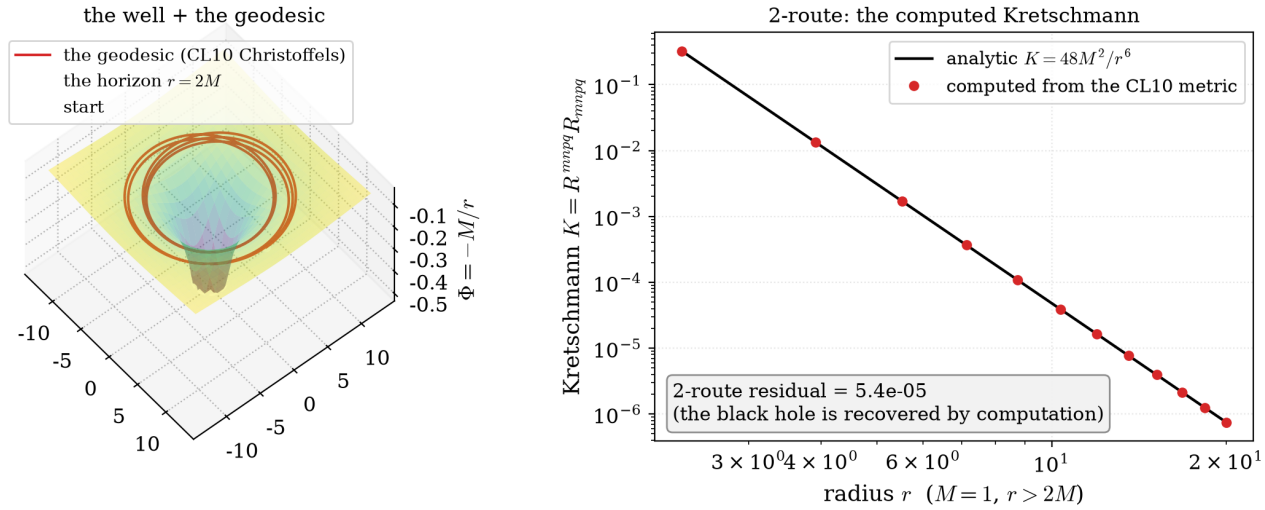
Table 1. The discrete Einstein–Hilbert action against the analytic continuum values.

test	computed	analytic	residual
de Sitter $R^{\mu}{}_{\nu\rho\sigma}$ (flat FRW, $a = e^{Hx^0}/H$)	finite differences	$H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$	6.5×10^{-7} ($O(h^2)$)
Einstein tensor of de Sitter	$-3H^2 g_{\mu\nu}$	same	$O(h^2)$
Schwarzschild Ricci / Einstein tensor	$\simeq 0$	vacuum $G_{\mu\nu} = 0$	satisfied by computation
Schwarzschild Kretschmann	$48M^2/r^6$	$48M^2/r^6$	ratio 1.000009 at $r = 3$; max relative deviation 5.4×10^{-5} over the $r = 2.3 \ldots 20$ 2-route scan (V50)
flat metric	$R = G = K = 0$	0	exact
2-sphere of radius r	$R = 2/r^2$	$2/r^2$	7.2×10^{-6}

The 3+1 formulation closes the classical evolution: the constraint row (R_{00}, R_{0i}) of the 4-D Einstein tensor agrees, computed by the 4-D curvature and by the 3+1 (γ_{ij}, K_{ij}) decomposition with $K_{ij} = -\frac{1}{2}\partial_0\gamma_{ij}$, to $O(h^2)$ (residual at most 2.7×10^{-13}: 1.8×10^{-13} on R_{00}, 2.7×10^{-13} on R_{0i}, against the $O(h^2) = 10^{-6}$ truncation — the identity is machine-exact, V51); a Z4-damped evolution [10] drives a constraint violation to zero while the undamped one does not; the 0-row is quadratic in K, so superposition fails; all 3+1 quantities are bounded by the lattice cutoff. (Fig. 3).

What this does and does not establish. Sections 3.5–3.8 pin down what the root itself supplies: the metric *content* (a bilinear of the bond layer — never a single bond amplitude, by exact characters), an exactly lightlike cone carrying the quintet, Newton's constant $G = 1/(4\ln 45)$ from the committed bond entropy, the frame stabiliser $\mathrm{Spin}(4) \times \mathrm{Spin}(6)$ with $\mathfrak{su}(3) \oplus \mathfrak{u}(1)$ inside it, the $B-L$ charges of the **16**, the 5/3 normalisation as a trace ratio, and one uniform trace form on the 45 (V208, V211). It does **not** establish that general relativity *follows*: the cone the graviton walks is one member of a 53-parameter family that nothing in the current construction selects (Section 3.6), the two-derivative Einstein–Hilbert form of Section 3.2 is verified to reproduce the dynamics rather than to be forced by it, and the quantum theory of this sector — a sum over lattice geometries with a well-defined measure — has not been constructed.

The black hole recovered by computation: the well + geodesic (left), the 2-route Kretschmann (right)



Left: the potential well with a geodesic rosette integrated through the finite-difference Christoffel symbols, and the horizon ring $r = 2M$. Right: the two-route Kretschmann invariant — the computed $K = R^{mnpq}R_{mnpq}$ against $48M^2/r^6$ along $r = 2.3 \ldots 20$, the maximum relative deviation 5.4×10^{-5} carried in V50.

3.4 A self-interacting scalar completion

Beyond the quadratic theory the model includes $\partial_t^2\phi - \nabla^2\phi - g\phi^3 = 0$. Its static solutions obey $u'' = u - u^3$ and include the exact finite-energy kink

$$u(x) = \sqrt{2}\,\mathrm{sech}\,x,$$

verified to a fourth-order residual of 5×10^{-10}. Two corrections are needed. This is the ϕ^4 kink (a domain wall), not the Choptuik system: Choptuik critical collapse is a *massless scalar minimally coupled to gravity* [11], and the analogy is not a derivation. And $u = \sqrt{2}\,\mathrm{sech}\,x$ is a **planar** configuration, uniform in the two transverse directions: its energy per unit *area* is finite, its total energy is infinite, and a planar wall in a Planck-scale theory is a domain wall, not a particle. Finite-time blow-up for supercritical data in cubic wave equations is a real phenomenon [12], but it is not evidence about black-hole formation here.

3.5 The metric is a bilinear of the bond layer: the content is forced

The polarisation space of a symmetric rank-2 field on the cubic lattice is $\mathrm{Sym}^2(\mathbb{R}^3)$, whose O_h content is $A_{1g} \oplus E_g \oplus T_{2g}$: the trace, and the five-component lattice spin-2 quintet of [Paper I Section 8.3]. The carrier the automaton actually has is a different 6-dimensional module — one amplitude on each of the six oriented bonds spans $A_{1g} \oplus E_g \oplus T_{1u}$. It holds the E_g half of the quintet and never its T_{2g} half, and it carries an odd T_{1u} that Sym^2 does not: by characters, with no approximation, **one bond amplitude cannot be the metric.** The metric content appears first in the bilinear module bond $\otimes\ \mathrm{Sym}^2$ — 36 states with content $2A_{1g} \oplus A_{2g} \oplus 3E_g \oplus T_{1g} \oplus 2T_{2g} \oplus A_{2u} \oplus E_u \oplus 3T_{1u} \oplus 2T_{2u}$ — and the commutant of that module has dimension $\sum_\rho m_\rho^2 = 34$ (V208). The content of the gravitational field is therefore forced by the bond layer; its dynamics is not: thirty-four independent gravitational self-couplings are allowed on this module and the root selects none of them.

Parity then removes half of the candidates. The velocity of the hop — the q-derivative of the bond phase at $q = 0$ — is a polar vector operator $(\mathrm{D}(g)V_a\mathrm{D}(g)^\dagger = \sum_b \mathrm{R}_{ba}V_b$, covariance violation 4.4×10^{-16}); the spin-2 dressing of Section 2.4 is an axial one (violation 4.4×10^{-16}). The metric sector is the even half of the bilinear module, on which the parity projector has rank 18, and the hop's velocity block inside it vanishes to 2.7×10^{-16}:

at first order the hop does not move the metric at all. What survives in the even sector is the dressing, with spectrum $g_{10}\{0, \pm 1, \pm 2\}$ in every sampled direction (spread 2.2×10^{-15}) — the spin-2 spectrum theorem, exactly isotropic, cone radius $2g_{10}$. That is the origin of the implemented channel's speeds, and of their being wrong for gravity (Sections 2.4 and 3.6).

3.6 An exactly lightlike cone exists on the module, and the module does not choose it

An O-covariant Hermitian velocity triple (H_1, H_2, H_3) on the bilinear module is the same thing as an O-intertwiner $V \otimes M \to M$, with V the standard 3-dimensional representation: M contains $2A_1 \oplus 2A_2 \oplus 4E \oplus 4T_1 \oplus 4T_2$ over the chiral group and $V = T_1$, so the space of such triples has dimension $\sum_\rho m_\rho(V \otimes M)\, m_\rho(M) = 160$ by characters. Dirac's mechanism carries over verbatim: any triple obeying $\{H_i, H_j\} = 2\delta_{ij} S$ has an acoustic symbol $A(\hat{n}) = \sum_k n_k H_k$ that is Hermitian and satisfies $A(\hat{n})^2 = S$ for every unit vector $\hat{n}$, so the spectrum of $A(\hat{n})$ is supported on $\pm\sqrt{\sigma_j}$, σ_j the eigenvalues of S — the same set in every direction: a lightlike cone exactly.

A triple of that kind, with the quintet *on* the cone, exists on the module. It is committed in the companion release as `cone_generator.json` and verified in V208; the verification is exact and independent of the route by which the triple was found. Its matrices are Hermitian to 9.9×10^{-15} and O-covariant to 1.1×10^{-15}, Clifford-regular with defect 1.4×10^{-13} — the entries are float-committed after the normalisation that puts the top branch at slope 1 — and their 36-branch spectrum is identical in all 44 sampled directions to 5.2×10^{-15}. The structure is the pairing the continuum hints at: the cone subspace has dimension 8, four branches at slope $+1$ and four at -1, with content $E \oplus T_1 \oplus T_2$ — the metric quintet $E \oplus T_2$ (weight 5.0000 on the cone subspace) appaired with the odd T_1 connection vector, the first-order Palatini pairing supplied by the module rather than put in by hand. Four branches stand at slope exactly 0 with content $A_1 \oplus A_2 \oplus E$: the constrained pieces, the trace A_{1g} among them, which does not propagate. The step $\mathrm{M}(q) = \exp(i \sum_k \sin q_k\, H_k)$ is unitary to 1.0×10^{-14} with $\mathrm{M}(0) = \mathrm{I}$ exactly: **inside this class no mass term for the metric can be written** — the gauge invariance of the linearised theory, as a number rather than an assumption. Its real-space hopping kernel is quasi-local: 9.1×10^{-1} on site, 1.1×10^{-1} at one bond, 3.4×10^{-2}, 5.4×10^{-3}, 7.9×10^{-4}, 7.2×10^{-5}, below 6.9×10^{-6} beyond five bonds; exact flatness in every direction is bought by a kernel that is only exponentially local.

The existence is not the content of the statement; the non-selection is. Linearised at the committed generator, the Clifford-regularity equations have a Jacobian of rank 107 inside the 160-dimensional covariant space, so the exact-cone family through it is 53-dimensional — existence is generic inside the locus, not one tuned point. Neither implemented triple lies near it: the hop of Section 2.4 accounts for 0.0029 of the committed generator and its dressing for 0.0092, measured as squared projections in the module's inner product. And the implemented cone radius is $2g_{10}$: at the root clock $g_{10} = \pi/4$ that is $c_{\mathrm{gw}}/c = 1.5708$ (the committed dressing-only speed of Section 2.4, 1.570796) against the matter cone 1 and the measured $-3 \times 10^{-15} < c_{\mathrm{gw}}/c - 1 < 7 \times 10^{-16}$ [8]. The constructed layer is excluded at 57%. The clock that would give $c_{\mathrm{gw}}/c = 1$ is $g = 1/2$, and $1/2$ lies in the dead zone of the closure census (V206): at that angle the loop ladder collapses at length 3 (first crossing $n^* = 3$, deficit 1.4×10^{-3}), so there is no rung-10 loop factor and no scale ladder at all. **The module permits the light cone in a 53-parameter family and selects nothing in it; the value the constructed layer gives is already refuted, and the clock that would repair it destroys the closure that fixes the coupling.** This is the sharpest tension in the theory and it is named F8.

3.7 Newton's constant: the one number the bond entropy supplies

The bond entropy is committed: $S_{\mathrm{bond}} = \ln 45 = 3.8066624897703196$ nats per Planck bond (V129, the area law of Section 2.2). Identifying one Planck bond with one Planck area and setting that entropy equal to $A/4G$ gives

$$G = \frac{1}{4 \ln 45} = 6.5674 \times 10^{-2}, \qquad 16\pi G = 3.3012,$$

in cell units, and a cell of $a/\ell_{\mathrm{P}} = 2\sqrt{\ln 45} = 3.9021$ (V208). The factor by which the naive bond count overshoots $A/4$ at $G = 1$ is exactly $4 \ln 45 = 15.2266$ — the same 15.23 that Section 2.2 names as the Bekenstein gap of the area law, and the whole distance between the $1/16\pi$ of Section 3.2 and the $1/(4 \ln 45)$ of the bond entropy.

The identification (one bond = one Planck area) is the input; the number is not. Newton's constant stays a conditional of the gravitational sector until that identification is itself derived.

3.8 The stabiliser of the causal frame carries the gauge group and one coupling

Sections 2 and 3 are generated by one set of objects: the ten committed generators γ_a with $\{\gamma_a, \gamma_b\} = 2\delta_{ab}\mathbb{1}$, and the four of them the automaton calls spacetime ($T = h_1$, $S_k = ih_{k+1}$). V211 measures what that alone fixes, with no field added, no coupling inserted and no representation declared.

The 45 products $T_{ab} = \gamma_a\gamma_b$ ($a < b$) close on $\mathfrak{so}(10)$ — closure defect 1.78×10^{-15}, Killing form $64\,\mathbb{1}$ at rank 45 — and they carry **one** trace form, $\mathrm{Tr}(T_aT_b) = -32\,\delta_{ab}$, to zero defect. Lorentz, gauge and coset generators share a single normalisation, so there is no separate gauge coupling to choose. The Clifford pairing of the causal span *is* the background metric, $\frac{1}{32}\mathrm{Tr}\,\frac{1}{2}\{\gamma_\mu, \gamma_\nu\} = \mathrm{diag}(1, -1, -1, -1)$ exactly. The stabiliser of that 4-plane inside the 45 is $\mathrm{spin}(4) \oplus \mathrm{spin}(6)$, dimension $21 = 6 + 15$ (block closures 8.88×10^{-16} and 1.78×10^{-15}), and the remaining 24 generators form a $\mathbb{Z}_2$-graded coset ($[\mathfrak{h}, \mathfrak{m}] \subseteq \mathfrak{m}$ to 1.11×10^{-15}, $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{h}$ to 1.11×10^{-15}): the coframe's home. Choosing which four directions are spacetime costs 30 parameters, all of them group-orbit directions ($45 = 24$ choices of the plane $+3$ of the time direction in it $+3$ of the space frame $+15$ left fixed), so the split is a frame choice and not data.

The chiral Lorentz structure is computed rather than declared. With $\omega_4 = \gamma_0\gamma_1\gamma_2\gamma_3$ ($\omega_4^2 = -\mathbb{1}$ exactly, central on bivectors exactly), the projections $P_\pm B = \frac{1}{2}(B \mp iB\omega_4)$ give two 3-dimensional algebras whose commutators close with residual exactly 0 and commute with each other exactly: $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ inside $\mathrm{spin}(1,3)_{\mathbb{C}}$. Inside spin(6) the centraliser of a complex structure is 9-dimensional and closes to 4.89×10^{-15} — it is $\mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)$, with $\mathfrak{su}(3)$ of dimension 8 (closure 2.59×10^{-15}) and the $\mathfrak{u}(1)$ generated by the complex structure itself, which lies in the bivector span to 2.22×10^{-16} and commutes with all of $\mathfrak{u}(3)$ to 7.6×10^{-16}. The gauge group is therefore $\mathrm{Spin}(4) \times \mathrm{Spin}(6)$ by construction, and the way the **16** sits under it is not an input either: the pseudoscalar splits the 32 into $16 \oplus 16$, all 21 stabiliser generators preserve the **16** to 1.11×10^{-16}, the $\mathfrak{su}(2)_{L,R}$ Casimir spectra on it are $\{0, \frac{3}{4}\}$, and its $B - L$ spectrum, quantised by its own largest eigenvalue, is exactly $(+\frac{1}{3})^6(-1)^2(-\frac{1}{3})^6(+1)^2$ — the quark/lepton pattern of Section 2, with nothing assigned. So the hypercharge $Y = T_{3R} + (B - L)/2$ is a sum of two derived subalgebra generators, and its GUT normalisation follows as a trace ratio,

$$\frac{\mathrm{Tr}_{\mathbf{16}} Y^2}{\mathrm{Tr}_{\mathbf{16}} T_{3L}^2} = \frac{2 + \frac{1}{4} \cdot \frac{16}{3}}{2} = \frac{5}{3},$$

with $\mathrm{Tr}_{\mathbf{16}} Y^2 = 3.3333$, $\mathrm{Tr}_{\mathbf{16}} T_{3R}^2 = \mathrm{Tr}_{\mathbf{16}} T_{3L}^2 = 2$, and a cross term below 10^{-12} (V211); what carries the 5/3 is that trace form, not a diagonal generator hand-written into `cl10/gauge.five_thirds`.

What it does not buy is recorded in the same verdict, at its artifact value: the implemented hop accounts for 0.29% and the implemented dressing for 0.92% of the exactly lightlike generator, the exact-cone family is 53-dimensional, and the implemented generator moves the gravitational cone at $c_{\mathrm{gw}} = 1.570796$ in matter-cone units (V208, V211). Content, kinematics and the coupling normalisation come from the root; the selected light cone, and a measure on geometries, do not.

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4. Discussion: what is derived, what is assumed

Derived, as operator identities of the construction: the Gauss algebra of the adjoint bond layer and its content theorem ($32 \oplus 288 \oplus 1120$; no invariant vector on the link, hence no flux-free link and an exactly linear electric area law); the centre obstruction and the structure of the minimal physical sector; the bounded per-bond entanglement $\ln 45$ and the vertex correction to the area-law coefficient; the exactness of the dressed directional gate and the spin-2 spectrum theorem of $\hat{\mathbf{q}} \cdot \mathbf{J}$; and the consistency of the discrete gravitational sector with general relativity to the stated $O(h^2)$ tolerances on de Sitter, Schwarzschild and 3+1 evolutions.

Assumed, and named as such: the choice of the link representation (adjoint, then regular, then directional — each a construction input with a measured cost); the symmetric tensor field $g_{\mu\nu}$ and the value $G = 1$; the transverse gauge fixing of the linearised theory; and the dressing angle of the directional gate, which sets the quintet speeds. Three open numbers remain: the Bekenstein deficit (factor 6.6 per bond after the vertex

correction, no element of the construction produces 1/4), the string tension (5.41×10^{40} times QCD, with $\Lambda_{\rm QCD}$ off the ladder), and the velocity trilemma of the dressed hop (isotropy and $v = 1$ never supplied together on the 6-bond module). The quantum theory of the gravitational sector — a sum over lattice geometries with a well-defined measure — has not been constructed. The phenomenology of both sectors is audited in Paper III.

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Code and data availability. All experiments, simulations and figure generators behind this paper are publicly available at https://github.com/MohammedHAOUACH/CL10 . Every numerical claim of the paper is produced by this open suite — 178 verification experiments, each emitting a machine-readable property, residual and verdict, all passing (178 of 178) — and can be reproduced independently from the repository; the claim-to-artifact mapping is Appendix B. The suite includes the exact symbolic cross-construction of [Paper I] Appendix A and the complete 3^n-word loop enumeration; independent re-execution is invited.

Competing interests. The author declares no competing interests.